

ZEROTRUST - ONE

Next-Generation SOC Operations Center & Payload Threat Inspector Workbench

System Architecture, Multi-LLM Consensus Engine, Payload Parsers & Security Specification Document

ENV: LIVE-SERVER

LLM CONSENSUS V3.0

STATUS: ACTIVE

1. EXECUTIVE SUMMARY

ZeroTrust-One is an enterprise-grade Security Operations Center (SOC) telemetry and payload threat inspection platform. Engineered with a zero-trust architecture philosophy, the system isolates, extracts, parses, and evaluates unknown or untrusted digital artifacts before they reach internal networks, mail servers, or endpoint environments.

By combining client-side multi-format extraction (PDF, QR Codes, Raw Buffer Headers) with a dual-LLM AI Consensus Engine (**Groq Llama-3.3-70b** and **Gemini-Flash**), ZeroTrust-One delivers real-time threat detection, heuristic risk scoring (0-100), and automated enforcement actions with ultra-low latency.

2. SYSTEM TOPOLOGY & ARCHITECTURE

2 AI Models

GROQ LLAMA 3.3 + GEMINI FLASH

< 750 ms

AVERAGE PIPELINE LATENCY

Core Architectural Components

- **Inference Node:** Handles payload dispatching, request normalization, and real-time execution telemetry streaming.
- **Client-Side Extractor Engine:** Leverages client-side PDF.js and jsQR visual decoding matrix algorithms to parse files locally before server submission, minimizing bandwidth and data exposure.
- **Consensus Matrix Engine (v3.0):** Runs parallel evaluation against dual large language models. Computes individual model risk scores and aggregates them into a final convergence verdict.
- **Heuristic Engine (Rule-v1):** Evaluates rule-based vector indicators alongside AI model predictions to prevent false negatives and prompt injection bypasses.

3. SUPPORTED PAYLOAD TYPES & CONTEXT MATRIX

ZeroTrust-One features specialized parser profiles and contextual inspection logic tailored for 6 distinct target payload vectors:

PAYLOAD TYPE	PARSER ENGINE	INSPECTION SCOPE & SECURITY FOCUS
RAW_EMAIL_HEADERS_AND_BODY	Strict Security Zero-Trust	Parses raw MIME/RFC822 structures, SPF/DKIM/DMARC authentication records, header anomalies, reply-to mismatches, and spear-phishing syntax.
HTTP_URL_STRING	URI Inspector Engine	

PAYLOAD TYPE	PARSER ENGINE	INSPECTION SCOPE & SECURITY FOCUS
UNSTRUCTURED_TEXT_PAYLOAD	NLP Heuristic Scanner	Evaluates URIs for typosquatting, IDN homograph spoofing, suspicious query string parameters, multi-stage redirect chains, and known domain risk metrics.
		Analyzes free-form text buffers for social engineering indicators, credential harvesting prompts, sensitive PII leakage, and jailbreak / prompt injection attacks.
		Decodes 2D barcode matrices to reveal embedded payload destinations, protecting against quishing (QR phishing) and obfuscated redirect exploits.
		Extracts text layers and structural object graphs from PDF files to scan for malicious JavaScript streams, embedded executable objects, and phishing attachments.
		Performs static telemetry extraction on binary artifacts, analyzing entropy, header signatures, structural section flags, and malicious file markers.
QR_CODE_IMAGE_SCAN	jsQR Visual Matrix	
PDF_DOCUMENT_PAYLOAD	PDF.js Object Extractor	
BINARY_FILE_ARTIFACT	Static Artifact Analyzer	

4. USER INPUT CONTEXT ENRICHMENT

Context Box Integration: When analysts select a target payload type or upload file artifacts, ZeroTrust-One presents an interactive User Context Field. This allows analysts to attach operational notes, origin data, or specific focus instructions (e.g. *"Inspect page 2 table calculations"* or *"Target payload suspected spoof of HR portal"*), which are injected into the multi-LLM prompt pipeline for enhanced detection precision.

5. MODEL CONSENSUS BREAKDOWN & SCORING MATRIX

Every scan produces an **Aggregate Risk Score (0 / 100)** derived from individual model evaluation nodes:

ENGINE NODE	MODEL NAME	OUTPUT SPECIFICATION	DEFAULT SCORE FORMAT
Groq Node	Llama-3.3-70b	High-speed structural analysis & threat pattern recognition	0 / 100
Gemini Node	Gemini-Flash	Deep contextual reasoning, multimodal analysis & threat verification	0 / 100

Verdict Threshold Spectrum

- **0 – 30 (LOW / SAFE):** No threat detected. Automatic pass / enforcement recommendation: **No Threat Detected.**
- **31 – 70 (MEDIUM / SUSPICIOUS):** Potential anomaly or low-confidence risk. Enforcement recommendation: **Quarantine & Review.**
- **71 – 100 (HIGH / CRITICAL):** Confirmed threat indicator or malicious payload. Enforcement recommendation: **Block & Terminate Session.**

6. API INTEGRATION ENDPOINT

```
POST /api/v1/scan HTTP/1.1
Content-Type: application/json

{
  "type": "RAW_EMAIL_HEADERS_AND_BODY",
  "payload": "Received: from mail.attacker.com ...",
  "context": "Reported by finance department user."
}
```